

# Arunabha Majumder

**Mechanical Engineer | New Product Development | Mechatronic Systems | SolidWorks**

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## PROFESSIONAL SUMMARY

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Mechanical Engineer with a Master's in Mechatronics and about three years of hands-on new product development at Aalborg University, taking electromechanical devices from concept through SolidWorks CAD, FEA, prototype fabrication, assembly and testing. Designed a hybrid-actuated shoulder exoskeleton built from machined and additively manufactured parts that won the Best Paper Award at IFToMM ISRM 2026. Certified SolidWorks Associate (CSWA) with a structured, analytical approach to problem solving and experience collaborating across mechanical, electronics, controls and biomechanics disciplines in international teams in Denmark and India.

## CORE SKILLS

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- **CAD & Simulation:** SolidWorks (CSWA certified), Fusion 360, Finite Element Analysis (FEA), MATLAB/Simulink
- **Design & Manufacturing:** New Product Development (NPD), Mechanical Design, Design for Assembly (DFA), Additive Manufacturing / 3D Printing, Machined Parts, Prototype Fabrication, Assembly, Custom Test Rigs
- **Mechatronic Systems:** Actuator Design, Motor / Encoder / Force-Sensor Integration, DAQ, Embedded Control, Pneumatic Systems, Control Systems
- **Analysis & Testing:** Stiffness and Kinematic Modelling, Experimental Validation, Test Planning, Data Analysis, Motion Capture (Qualisys), EMG
- **Communication:** English (C1, fluent written and spoken), cross-disciplinary collaboration, ownership from concept to tested hardware

## PROFESSIONAL EXPERIENCE

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**Research Engineer – Robotics & Actuator Development (PhD Project: VIEXO) | 2023 – Present**  
*Aalborg University, Aalborg, Denmark*

- Owned the full product development cycle of a wearable robotic device from concept to tested hardware: SolidWorks CAD and FEA, mechanical fabrication, assembly, motor/encoder/force-sensor and DAQ integration, embedded control and custom automated test rigs.
- Cut shoulder exoskeleton energy consumption by more than 25% by designing a hybrid actuation architecture that combines a parallel spring with a 6 Nm motor to offload gravity torque; awarded Best Paper, IFToMM ISRM 2026.
- Removed the need for an external force/torque sensor in a 5-bar parallel robot by deriving a sensorless force-estimation method from the actuator torque–deflection characteristic, reducing system cost, mass and integration complexity.
- Validated analytical stiffness models against experimental hardware to below 10% RMS error by mapping variable-stiffness actuator (VSA) states into Cartesian workspace stiffness, de-risking the mechanism before fabrication.
- Reduced user anterior deltoid muscle effort by more than 15% by deploying an Assist-as-Needed hybrid controller on functional prototype hardware, validated on human subjects.
- Planned and ran a 12-participant validation study across 3 assistance modes and 2 load conditions, covering instrumentation (EMG, motion capture), ethics protocol and the end-to-end data pipeline; results support two journal submissions.

**M.Tech Researcher – Actuator Systems (Master's Thesis) | 2020 – 2022**

*CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India*

- Reduced dynamic position-tracking error to 0.3–0.78% by designing a neural-network-based gain-scheduled PID controller for a non-linear pneumatic actuator, outperforming the classical PID baseline under varying axial load.

- Fabricated pneumatic artificial muscle actuators in-house and integrated valves, pressure sensors and DAQ to characterise load–displacement behaviour and derive an empirical actuator model.

### **Mechanical Engineering Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019**

*Siddaganga Institute of Technology, Tumakuru, India*

- Led CAD design, fabrication and user testing of an upper-limb rehabilitation exoskeleton; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant. Demo: [youtube.com/watch?v=q5Ystz9NCpQ](https://youtube.com/watch?v=q5Ystz9NCpQ)

## **EDUCATION**

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### **Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present**

*Aalborg University, Department of Materials and Production, Aalborg, Denmark*

### **M.Tech, Mechatronics | 2020 – 2022**

*AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India*

### **B.Tech, Mechanical Engineering | 2015 – 2019**

*Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India*

## **ENGINEERING PROJECTS**

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- **Go-Kart (2017):** Led end-to-end design and fabrication of a custom Go-Kart built from the ground up.
- **Efficycle (2018):** Worked in a multidisciplinary team to engineer a human-electric hybrid vehicle for the Efficycle competition.

## **CERTIFICATIONS & AWARDS**

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- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- Best Major Project Award, Department of Mechanical Engineering (2019)
- TATA Technologies Innovation Award – INR 100,000 Grant (2019)

## **PUBLICATIONS**

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- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026, MMS*, vol. 213, Springer. (Best Research Paper Award)
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

## **TECHNICAL TOOLS**

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- **Software & Programming:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino, LaTeX
- **CAD & Manufacturing:** SolidWorks, Fusion 360, FEA, Bambu Lab and Ultimaker Cura (3D printing)
- **Instrumentation & Testing:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, Pneumatic Control, Custom Rig Fabrication
- **Visual Design:** Adobe Illustrator, Adobe Photoshop; Diploma in Fine Arts (Painting), Bangiya Sangeet Parishad – sketching and technical illustration

## **LANGUAGES**

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English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

## **REFERENCES**

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Available upon request.